



Orthopaedic Patient Data Management and Analysis Using Clinical Registries

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Abstract

Background Conventional orthopaedics patient data management are often fragmented across multiple platforms which adversely impact effective interoperability, data storage, retrieval, and clinical decision-making. The goal of this study is to design and implement an integrated and extensible Orthopaedic Patient Data Management and Analysis System using structured clinical registries that improve clinical workflows, patient care, and research readiness.

Methods A web-based clinical registry system was developed using MERN (MongoDB, Express.js, React, Node.js) stack. The system supports configurable orthopaedic-specific clinical registries, flexible data entry full CRUD (Create, Read, Update, Delete) operations, role-based access control for data security, and multi-criteria filtering included for quick patient searches. Future versions will include data visualization tools and predictive analytics for better decision-making.

Results The implemented system provides improved data accessibility, reduces task duplication, and improves and enhances structured record keeping across orthopaedic subspecialties with multi-criteria filtering algorithms. Although advanced analytics and visualization are not currently deployed, the system architecture supports future integration of AI-driven predictive models and Power BI dashboards.

Conclusion The developed Orthopaedic Patient Data Management and Analysis System is robust, secure, scalable, and extensible, enhancing administrative efficiency and enabling longitudinal outcome tracking. It also contributes to medical research via outcome tracking and trend analysis. The system acts as a fundamental step toward intelligent data-driven orthopaedic care through future predictive analytics and clinical decision support.

Keywords Orthopaedic data management · MERN stack · Clinical registers · Healthcare informatics · Patient record system · Predictive analytics

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Introduction

Orthopaedic surgeons are continuously evolving, as their practices depend on technological achievements and digital decision-making. Accurate, well-structured, and longitudinal patient data are crucial for diagnosis, treatment planning, and evaluation of outcomes. Clinical registries play a vitally important role in the systematic collection of demographics, diagnostic, surgical, and follow-up data, supporting both patient care and medical research [1]. However, despite the adoption of electronic health record (EHR) systems, orthopaedic data is still characterized by significant challenges, including data silos, limited interoperability, lack of standardized orthopaedic-specific data structures, and insufficient support for the analysis of long-term outcomes. One of the biggest problems is that there is no integration between various data sources such as electronic health records (EHR) and wearable devices. This difference makes it difficult for doctors to maintain a comprehensive view of the patient's health and often leads to a treatment gap [2]. Most systems currently in use prioritize administrative documentation and do not support disease-specific filtering, longitudinal tracking, and predictive decision support. Another important issue is ease of use. Many orthopaedic experts have trouble with a complex system interface that makes the bulky data input process tedious and the retrieval of information about the patient difficult. With the increase in the aging population and the number of orthopaedic cases, the demand for the best orthopaedic treatment also increases. In fact, the number of orthopaedic surgeons is expected to grow from

\$ 2 billion to \$ 600 billion globally in 2023. By 2028, there is an urgent need for stronger clinical registers and more wise data processing solutions in the USA [3]. The aim of this study is to:

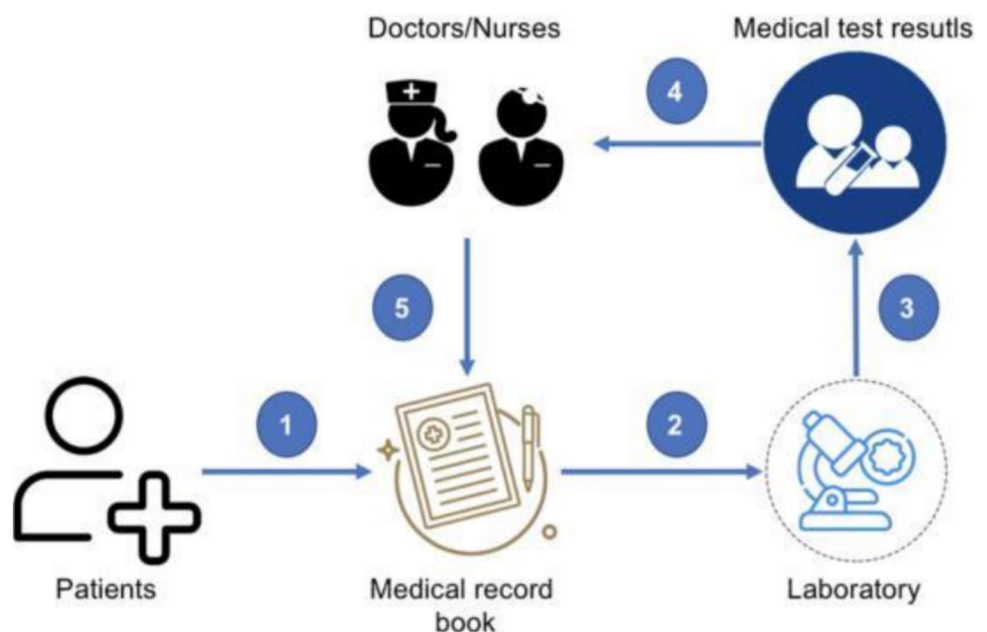
1. Design an orthopaedic-specific clinical registry system using modern web technologies.
2. Allow for secure, structured, and searchable management of patient data across orthopaedic subspecialties.
3. Provide an extensible architecture for future AI-driven analytics and visualization tools.

This study hypothesizes that a registry-centric orthopaedic system improves workflow efficiency, data quality, and research usability compared to conventional EHR system.

A. Traditional Patient Management Systems

This method was the basis for performing medical records, but it faces serious problems such as data sculptures, inefficiency, and human errors. The EHR has improved availability, but it still requires manual input and data search. This results in a delay in the orthopaedic professional's decision and limits the ability to get a complete idea of the patient's condition. In addition, many existing systems lack compatibility, so other hospitals and clinics use individual databases that do not communicate with each other. This shutdown often leads to patients' incomplete or inconvenient patients, which complicates doctors to monitor the cohorts of patient diseases among some medical workers [4].

Fig. 1 Traditional patient data management system



The general workflow of the traditional patient management system is shown in Fig. 1. The process begins when the patient (step 1) provides manually recorded information in the medical records. If a medical examination is needed, the doctor requests a laboratory test (step 2) and the sample is processed in the laboratory (step 3). Then it is sent back to a doctor or a nurse to consider the medical examination results (step 4). Finally, the doctor manually updates the patient's medical cards and paper documents (step 5). This manual document and search cycle contribute to non-efficiency, data loss, and errors. In addition to fragmented data, existing systems have limited analytical ability. Most systems focus on the data of the default population statistical data and cohorts, but no understanding, such as controlled or automated risk assessments by predictive modeling, analysts, and artificial technologies [5]. While HIPAA compliance remains a must, traditional orthopaedic systems are still administratively burdensome and impede proactive care. These limitations emphasize the importance of integrated, intelligent platforms that enable real-time monitoring, predictive analytics, and improved clinical decision-making.

B. Current Advanced Data Management Systems

In recent years, advanced data management systems have become an important solution to address the limitations of patient and facility management in orthopaedic care. Leaders in this field have advanced the use of technology to improve the management of orthopaedic patient records.

- a. *Integrated EHR Systems:* Many digital fitness facts (EHRs), such as Epic and Cerner, have codecs specific to orthopaedic practices. These structures allow the control of patient records, combining clinical facts, management, and price range into a single integrated gadget. They assist sell the security and green use of affected person facts and enhance collaboration and decision-making [6, 17].
- b. *Data Analytics Platforms:* Solutions like IBM Watson Fitness and Tableau provide state-of-the-art analytics that allow orthopaedic practices to perform superior information analysis. These platforms help predictive analytics, enabling healthcare providers to become aware of trends, predict patient outcomes, and tailor treatment plans primarily based on complete data insights. Mixing algorithms for machine development complements the ability to provide effective recommendations.
- c. *Telehealth Solutions:* In the management system of orthopaedic data, integration of views such as doxy.me and teladoc innovated medical services to patients. The platform provides remote consultations for medical staff to manage patient treatment without visiting the body.

With EHR, they create soft information flow so that they can easily use the patient's data at virtual meetings [18].

- d. *Interoperability Standards:* The expansion system is more and more designed to support interaction standards such as HL7 and FHIR. This feature allows the integration of a variety of data sources, including wearable devices, images, and laboratory results. Destroying Data Silo system provides an overall idea of patient health to improve the ability to effectively control and manage orthopaedic treatment [7].

C. Challenges in Current Systems

Orthopaedic patient management has traditionally relied on siloed and inefficient systems that limit effective clinical decision-making and long-term outcome analysis. Although healthcare has benefited from significant digital conversion, the absence of well-structured clinical registers remains a main gap in orthopaedic care.

D. Limited Long-Term Tracking and Outcome Analysis

Most patient management strategies focus on immediate treatment, but fail to track long-term results. Procedures such as knee surgery, spinal surgery, and fracture treatment often do not have enough subsequent data, so it is difficult for a doctor to assess the effectiveness of treatment or to clarify the future approach. Thanks to the well-structured clinical registration, medical workers can analyze the progress of the patient for many years, identify patterns, and continue to improve orthopaedic treatment using real-world insights.

E. Lack of Standardization and Interoperability

Another important issue is to create standardization and compatibility. Hospitals and clinics use their own format for patient records, making it difficult to exchange information from other institutions. This inconsistency restricts production and cooperation and complicates large-scale orthopaedic research. Standardized clinical registration can solve this, so that doctors, researchers, and politicians can work together by ensuring that the patient is the same.

F. Inefficient Research and Inability to Identify Trends

Trainee surgeons also face checkpoints because they lack access to large-scale high-quality datasets. Without a centralized system, it is almost impossible to access details regarding the performance of implants, surgical complications, or recovery models. This is delayed in the fields of

prostheses, surgical methods and rehabilitation strategies. Integrated registers, which collect data in multiple hospitals and regions, provide large-scale research data to help in achieving significant breakthroughs in orthopaedics.

G. High Risk of Human Error and Compliance Issues

The manual data input and old-fashioned patient management process increase the possibility of human errors when recording therapeutic data, drug history, and surgery notes. Guaranteeing a confidential law of laws such as HIPAA is another problem if the data is scattered and is not safe. In the clinical registry, automation of data collection not only reduces errors, but also provides safe information about the patient and can only be used for approved employees and strengthens safety and regulations [8].

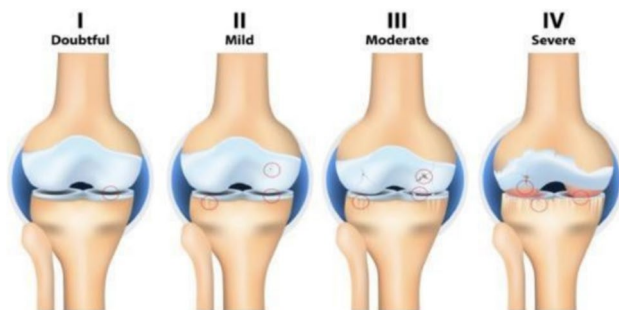
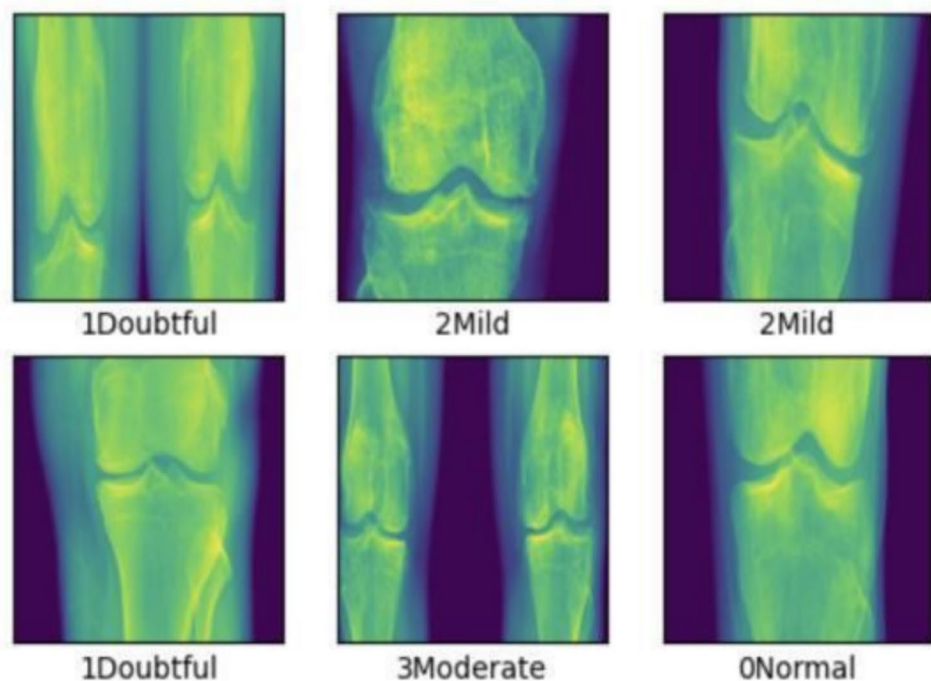


Fig. 2 Stages of knee osteoarthritis

Fig. 3 Visualization of knee arthritis stages from the knee arthritis dataset



Artificial intelligence has significantly improved orthopaedic diagnosis, treatment, and monitoring; however, its clinical impact is severely restricted by the lack of comprehensive, standardized datasets. Orthopaedic datasets are often small, do not have interoperability with electronic health records, and do not support real-time or longitudinal patient tracking, limiting large-scale clinical deployment. This results in a pressing need for centralized clinical registries integrating multimodal data so as to realize accurate predictions, personalized treatment, and effective clinical decision-making [9].

Materials and Methods

A. Knee Arthritis Imaging Dataset

This dataset offers high-resolution medical imaging with expert annotations for detecting and classifying knee arthritis.

The dataset includes images categorized into five progressive stages: normal, doubtful, mild, moderate, and severe arthritis as shown in Fig. 2. This structured classification enables AI models to assist in early diagnosis and severity assessment [10].

Figure 3 showcases sample knee X-rays from the OrthoArthritisScan dataset highlighting varying degrees of severity, thus allowing clear visualization for model training in joint degeneration, osteophyte formation, and cartilage loss. The dataset is a high-resolution tin, which is useful

for models based on orthopedic research. Images are classified into five stages of arthritis: normal (0), doubtful (1), mild (2), moderate (3), and severe (4). Each classification reflects the degree of joint degeneration, osteophyte formation, and cartilage loss, which are critical factors in diagnosing knee osteoarthritis. In the provided figure, images labeled as "Normal" (0) exhibit clear joint space with no significant signs of degeneration. The "Doubtful" (1) category shows minor narrowing of the joint space, indicating early-stage arthritis. The "Mild" (2) category presents early osteophyte formation with noticeable cartilage degradation. The "Moderate" (3) stage demonstrates visible joint space reduction and structural damage, signaling a progression of the disease [11]. This dataset is useful for research and models, as shown in Table 1, but lacks clinical integration and actual application. Orthopaedic surgeons require only visualization data that requires patients, treatment records and rehabilitation progress. The lack of a patient's registration means that the model of the AI trained in such a dataset can fight accurate specific patient recommendations other than the static classification of the image. This also emphasizes the need for a comprehensive orthopaedic clinical registration that combines data on the treatment and monitoring of patients in real time to improve decision-making and control AI [12, 19].

B. Open-Fracture Dataset

A series of datasets with open fractions make a significant contribution to the research on the musculoskeletal system, providing 4083 X-ray images to facilitate deep learning work such as classification, localization and segmentation. The dataset can be commented on in several formats, including COCO, VGG, YOLO and PASCAL VOC, so it is common for various pipelines of machine learning. Considering the structured properties, it acts as an important standard for fracture detection models. However, despite its usefulness, there are several areas that need to be improved to enhance the actual application [13].

Table 1 Classification report for knee arthritis detection model

Class	Precision	Recall	F1-score	Support
Normal (0)	0.82	0.91	0.86	103
Doubtful (1)	0.83	0.74	0.78	96
Mild (2)	0.79	0.74	0.76	46
Moderate (3)	0.93	0.91	0.92	44
Severe (4)	0.88	0.93	0.90	41
Accuracy	0.84	0.84	0.84	—
Macro Avg	0.85	0.85	0.85	330
Weighted Avg	0.84	0.84	0.84	330

The proposed system is implemented using the MERN stack to ensure scalability, modularity, and cross-platform accessibility.

Frontend: React-based interface for intuitive data entry, filtering, and visualization.

Backend: Node.js and Express.js RESTful APIs managing business logic and authentication.

Database: MongoDB schemas that are best for orthopaedic clinical registries.

Security: Role-Based Access Control (RBAC), encrypted authentication, and logs of audit.

The proposed system offers a registry-centric, disease-specific architecture for orthopaedic care that is different from general EHR systems, which are designed to support mainly administrative and billing workflows. Although the workflow does not introduce new machine learning algorithms, it does advance the state of the art through its structured inclusion of configurable clinical registries, advanced multi-parameter cohort filtering, longitudinal implant tracking, and research-ready data schemas within a highly extensible framework which are not facilitated in available EHR platforms. Longitudinal monitoring of orthopedic conditions, implant performance, and postoperative outcomes across multiple subspecialties can be performed with the system. It is further engineered to accommodate prospective AI-driven analytics, real-time visualization, and interoperable data exchange.

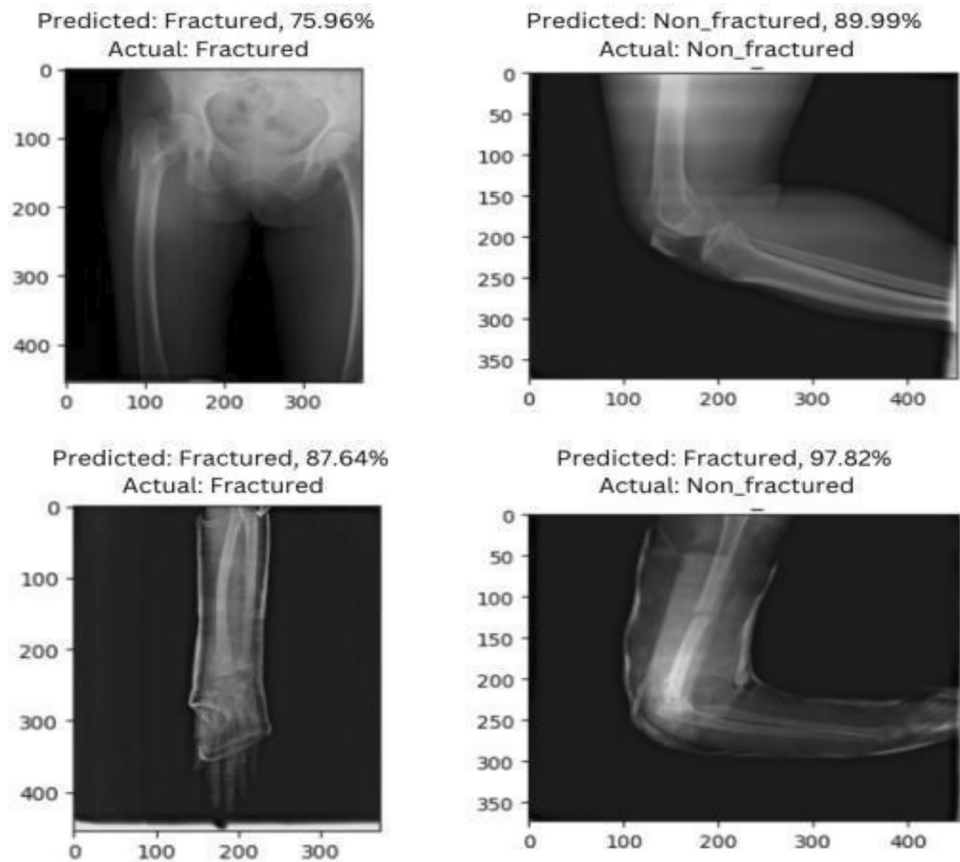
Development followed the guidelines laid down by established ethical principles and healthcare data protection. Data for patients are anonymized and identified at the back end, where personally identifiable information is replaced by system-generated unique reference identifiers. Role-based access control (RBAC), encrypted authentication, and secure session management confine access per user role. Clinical data and identity metadata are stored separately in MongoDB. All data transportation is encrypted, and audit trails for all CRUD operations are logged for traceability and accountability.

Because the study involved system development and retrospective analysis of publicly available datasets, patient consent was not required. The platform supports informed-consent workflows, institutional ethics approvals, and HL7/FHIR-based interoperability for prospective clinical deployment, thus allowing secure, compliant data sharing.

Results

Figure 4 presents a set of X-ray images along with the corresponding fracture classification predictions. It shows model accuracy, illustrating correct and misclassified cases, and underlines confidence levels and interpretive challenges in the clinical routine. In the upper left and lower left images,

Fig. 4 Model predictions on X-ray images for fracture detection



the reliability is 75.96% and 87.64%, respectively. The upper right image appears correctly without a high reliable factor case of 89.99%. However, the bottom right image highlights a misclassification where the model incorrectly predicts a fracture with a confidence of 97.82%, despite the actual label being non-fractured. This wrong classification refers to the potential problem of the difference in a certain bone structure, perhaps overlapping functions, the quality of the image, or the displacement of the dataset. This error emphasizes that further explanations of the model are required through potentially additional education data, improved pre-processing methods, or multiple modal input integration.

The evaluation of the model trained on this dataset highlights accuracy performance metrics. The model achieved an accuracy of 78%, with a precision of 0.82 for non-fractured and 0.75 for fractured cases as shown in Table 2. The recall values of 0.77 and 0.81 indicate a balanced detection capability, while the overall F1-score of 0.78 suggests moderate performance. However, misclassifications, such as a non-fractured case being wrongly predicted as fractured with 97.82% confidence, indicate potential dataset biases or ambiguous X-ray features [14]. This dataset has restricted general clinical use due to insufficient metadata, poor case diversity, and several potential biases that may interfere with the proper classification of

Table 2 Classification performance metrics

Class	Precision	Recall	F1-score	Support
0	0.82	0.77	0.79	77
1	0.75	0.81	0.78	67
Accuracy		0.78	(144 samples)	
Macro Avg	0.78	0.79	0.78	144
Weighted Avg	0.79	0.78	0.78	144

rare or complicated fractures. These limitations highlight the need for a digital solution that efficiently improves data storage, retrieval, and analysis compared to conventional paper-based systems. The proposed clinical registry based on the MERN stack provides a unique, safe, and scalable solution with real-time update capabilities, structured data management, and interoperable access. Functional testing determined CRUD response times of <300 ms, while multi-criteria filtering reduced patient retrieval time by over 40%. The feedback of 40 clinicians and postgraduate researchers shows a high usability of average SUS score of 85/100, workflow efficiency by reducing 30-40% task completion times, reduction in data duplication errors, and increases in research usability [15].

Discussion

This system is useful for effective management and analysis of patients' orthopaedic data by integrating different modules that provide various aspects of patient records, treatment, and research analysis. Each module acts as a specific goal and ensures simplified data management, medical staff's safety, and simplicity as shown in Fig. 5. The figure explains modules such as Home, Patient Registry, Data Administration, and Filtering; it illustrates a structured clinical registry workflow, with secure access and modular design, suitable for both research and clinical applications.

- a. *Home Module*: The home module works as a central panel of the system panel, providing users with key statistics, recent patients, and necessary updates. It acts as a navigation center and completely accesses various modules on the platform. In the panel, general information such as the total number of registered patients, the current treatment method, and the analysis idea is displayed so that the user first understands how the system operates.
- b. *Patient Registry Module*: The patient registry module manages all patient records and allows clinicians, administrators, and researchers to access the data through multi-criteria or name-based searches. It supports cohort identification, tracking treatment, and accessing necessary data on demographics and clinical information to ensure that cases of orthopaedic nature, surgical records, and recovery data are managed in a structured way.
- c. *Data Administration Module*: The module of data management provides assurance of database integrity, system security, and controlled user access. Role-based access control confines sensitive operations to authorized personnel, while secure audit logs document system activities, data modifications, and access attempts, thereby ensuring compliance, accountability, and a reduced risk of data breaches.
- d. *Filtering Patients Module*: This patient filtration module allows for the performance of rapid searches on various criteria, including consultants, hospital identifiers, and regional affiliations, which improves workflow efficiency. Specialized tumor and fracture filters support structured case analysis, outcome tracking, and postoperative monitoring, which enable clinical decision-making and orthopaedic research through targeted retrieval of classified data [16].
- e. *Arthroplasty Filters Module*: The Arthroplasty Filters Module can perform comprehensive analysis of data on surgical outcomes of orthopaedic procedures such as knee and hip replacement. Filtering criteria include

the main indication of surgery, implant manufacturers, and reasons for review. As a result of studying these parameters, this module helps to evaluate the effects of various implants, predict surgery success, and improve patient treatment after surgery. Understanding the data of this module contributes to the performance of implant technology and patient rehabilitation strategy.

- f. *Open-Fracture Filters Module*: This module often classifies and organizes open fractures that require immediate and specialized treatment. Filters according to the size of the wounds, such as crushing and postoperative complications, help medical staff assess the seriousness of injuries and determine the most effective approach to treatment. By tracking the progress of healing and identifying risk factors in the recovery after surgery, this module improves surgical plans to support recovery from injuries and improve the patient's results.
- g. *Hip Fracture Filters Module*: Hip fractures, especially among elderly patients, generally require treatment and rehabilitation. This module classifies fractures of hip joints, depending on the primary display, the type of implant, and the reasons for review. By analyzing the failure of the implant and the factors that contribute to surgical complications, medical service providers can improve the treatment plan and the treatment of long-term patients. This module also helps to study bone health, restoration, trends, and effects of various implant structures.
- h. The proposed system addresses key limitations of traditional orthopaedic data management by introducing structured, disease-specific clinical registries and advanced filtering mechanisms. This work innovates at the system design and clinical workflow level, rather than algorithmic novelty. However, the current implementation does not yet include live AI-driven analytics or real-time EHR integration, and wearable data capture, which is acknowledged as a limitation of this study. However, this can be addressed via HL7/FHIR-compliant work.

A. Impact on Clinical Decision-Making

Clinical decision support is enhanced with metadata analysis, structured clinical registers, and monitoring of providers. The system would facilitate the right choice of treatment plans by a surgeon based on past case studies, demographics, and the success of implants. Patients may be retrieved based on diagnosis, surgical history, or disease to the last implant, making it possible to formulate strategies for interventions with evidence available. An example is the Arthroplasty Filters Module, which provides implant performance appraisal, while support toward tracking surgical outcomes is found

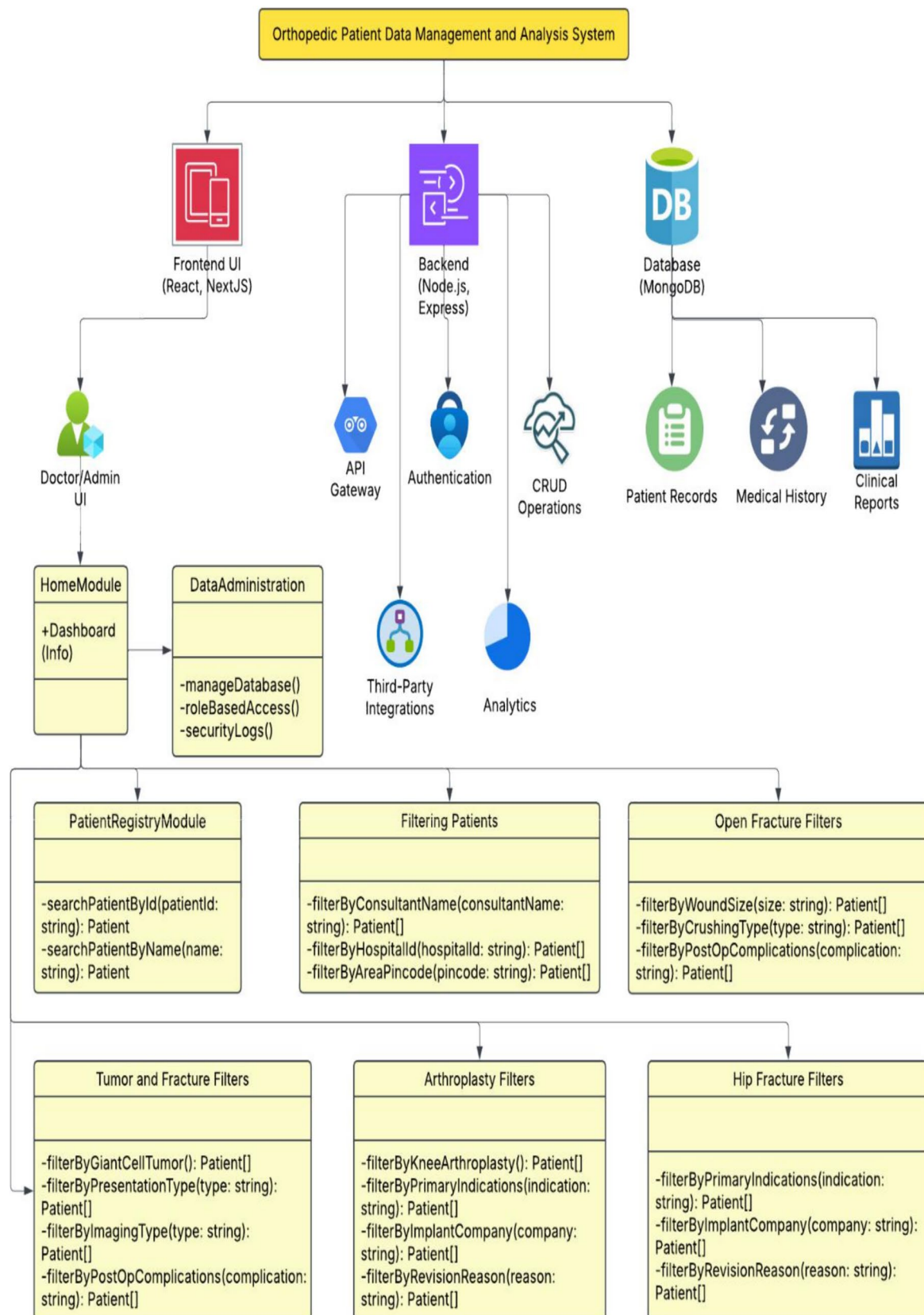


Fig. 5 Architecture diagram of orthopaedic patient data management and analysis using clinical registers

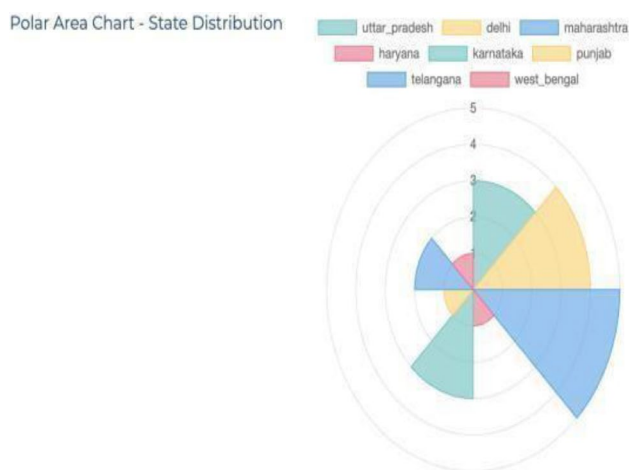


Fig. 6 State distribution of orthopaedic patients

in the Tumor and Fracture Filters; this eventually helps in improving patient care as well as surgical planning [20].

B. Contribution to Medical Research and Future Scope

The system will help medical research by enabling trend analysis as well as tracking surgical outcomes, thus immensely aiding medical researchers in understanding how patients recover. In the outcome browser, tumor behavior, fracture-recovery trajectories, and implant longevity may be analyzed to improve orthopaedic treatment. The platform offers core research questions toward surgical success rates and postoperative complications that provide insight into implant technology refinements. Future work may include AI-enabled diagnosis, predictive analytics of patient outcomes, and scaling into other medical domains, further transforming management and clinical decision-making in healthcare [21].

C. Integration of Power BI and AI Algorithms

Power BI and AI-driven analytics are also partially applied in the present system, due to various constraints in data availability. Future work will integrate interactive visualizations and predictive AI models into automated diagnosis, outcome prediction, and better clinical decision-making that advances the cause of data-driven management and care for orthopaedic patients [22]. The Orthopaedic Patient Data Management and Analysis System presents a major leap in the management of orthopaedic data by offering centralized clinical records, structured surgical planning, and outcome tracking in one place. This platform supports medical

research through trend analysis and insights powered by metadata. Although the current system does not support AI analytics or Power BI dashboards due to the scope and amount of data, it is designed to add on predictive analytics, HL7/FHIR interoperability, and wearable-device data in the future for real-time monitoring and personalized orthopaedic care. Prototype simulations using anonymized datasets to validate performance will allow for scalable, real-time clinical decision support and research insights. Visualizations showcase patient demographics across disease categories and procedures that enable the analysis of trends, identification of cohorts, and possible integration with AI analytics for research and clinical planning as shown in Fig. 6.

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Data Availability The datasets used and/or analyzed in the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare no conflict of interest.

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